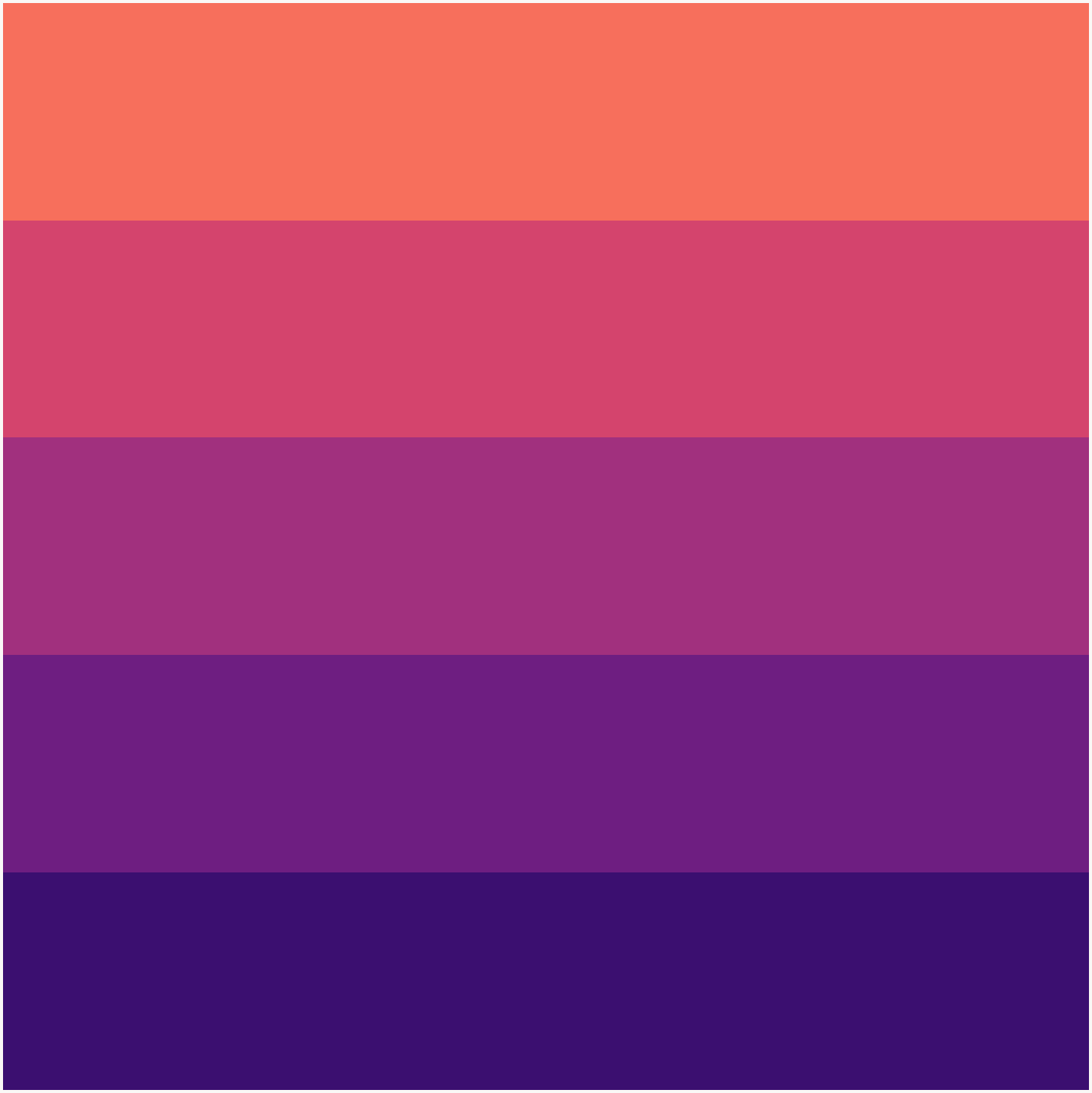
# Making Sense of Politics.



# Course handbook

POLI10302 · 20 credits

2026/2027

# Module cheatsheet

## Teaching team

### Dr Jack Bailey
Module leader

Mon, 13:30–15:30

Email · Book meeting

### Dr Kenn Rushworth
Workshop leader

Thu, 14:00–15:00

Email · Book meeting

### Cressida Arkwright
Workshop leader

To be confirmed

Email · Book meeting

## Assessments

### Assessment 1: Data journalism article
1,000 words ± 10% · 35% · [date and time]

- Analyse a YouGov tracker in R
- Create an original ggplot2 figure
- Cite at least two academic sources

Brief · Checklist · Support session

### Assessment 2: Experimental report
2,000 words ± 10% · 65% · [date and time]

- Analyse an experiment in R
- Include a table or figure
- Cite at least five academic sources

Brief · Checklist · Support session

## Weekly plan

- Do the reading
- Attend the lecture
- Attend the workshop
- Watch the bonus material
- Book into office hours if needed

## Help and support

**Questions:** jack.bailey@manchester.ac.uk

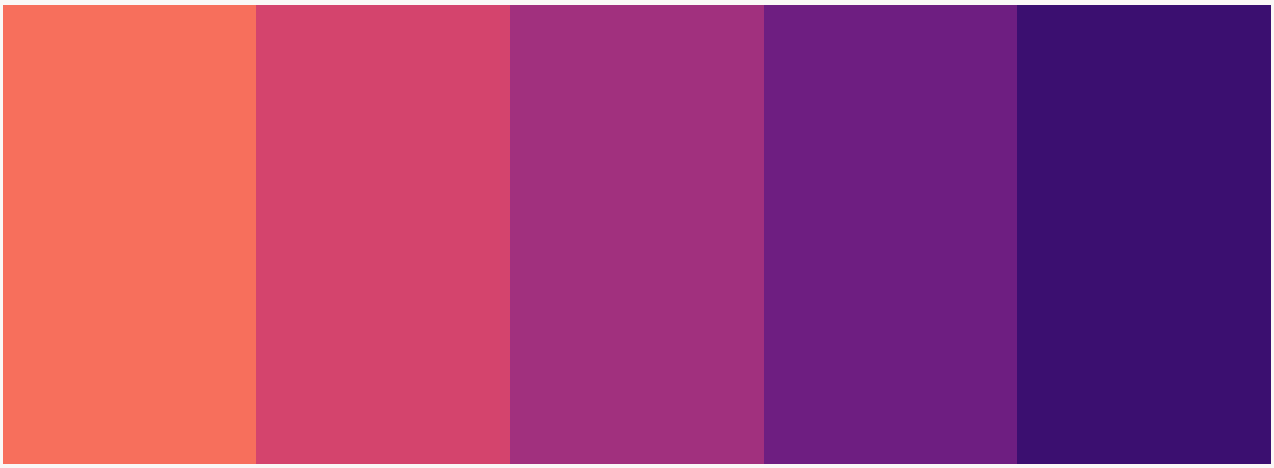
**Attendance:** soss.attendance@manchester.ac.uk

**Extensions:** soss.mitcircs@manchester.ac.uk

**Wellbeing:** soss.wellbeing@manchester.ac.uk

**Anything else:** soss.hub@manchester.ac.uk

# Welcome to Making Sense of Politics!



On this course, you'll learn to **analyse real political data** and **build data science skills** you can put on your CV. We'll **start from scratch**, so you don't need any experience with coding or data analysis. As you learn to **write code in R** and **make data visualisations using ggplot2**, you'll use your new skills to answer interesting political questions and explore how political scientists use experiments to tell cause from mere coincidence. Throughout the module, you'll learn how **professionals use these skills at work**, whether they work in research, politics, or industry. Then, in the final lecture, you'll have the chance to **ask a guest speaker** about the career paths that data science and statistics can open up once you finish your studies.

## The module

**Module cheatsheet** 2
Contacts and assessment essentials

**What you'll learn** 4
Questions, data, and career skills

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Tasks, deadlines, and support

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Skills to take into your career

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Lectures and workshops at a glance

**What's next?** 20
Free books to keep learning

## The weekly guide

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Week 3 Measurement and trends 11
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Week 5 Visualisation and data journalism 13
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# What you'll learn

### Answer questions

Turn your curiosity about politics into questions you can answer with data. You'll use evidence to explore patterns in public opinion and learn how experiments help us to work out what causes what.

### Learn data science

Find, prepare, and analyse data in R. You'll create stunning data visualisations with ggplot2, interpret statistical results, and think critically about what the data can — and cannot — tell us.

### Build useful skills

Practise asking questions, working with data, and explaining your findings. These are valuable skills, and give you concrete examples to draw on in applications and interviews.

# Assessments

## Assessment 1: Data journalism article

Investigate your own political question using a YouGov tracker. Analyse it in R, include an original ggplot2 figure, and write for a general audience.

| | |
|---|---|
| **Length** | 1,000 words ± 10%; reference list excluded |
| **Weight** | 35% of the module grade |
| **Deadline** | To be confirmed on Canvas |
| **Submit** | One PDF on Canvas |
| **Sources** | At least 2 academic journal articles or books |
| **Support** | Office hours and Workshop 5 in Week 6 |

Read the brief · Open the checklist

## Assessment 2: Experimental report

Develop a hypothesis from research and analyse an experimental dataset in R. Interpret the effect and uncertainty, using at least one table or figure.

| | |
|---|---|
| **Length** | 2,000 words ± 10%; reference list excluded |
| **Weight** | 65% of the module grade |
| **Deadline** | To be confirmed on Canvas |
| **Submit** | One PDF on Canvas |
| **Sources** | At least 5 academic journal articles or books |
| **Support** | Office hours and Workshop 10 in Week 11 |

Read the brief · Open the checklist

# Employability

**Build examples of what you can do, explain how you did it, and show why it matters.**

Working with evidence is useful in research, journalism, public policy, charities, and business. The module gives you concrete work to discuss in applications and interviews.

## Find and work with evidence

Locate data, assess its source, and prepare it for analysis. Keep a script that records your steps so you can explain and revisit them.

## Use judgement

Compare groups, recognise limitations, and distinguish correlation from causation. Interpret uncertainty alongside the size of a result.

## Communicate clearly

Choose an appropriate graphic, explain a finding in plain language, and adapt your writing to the audience.

### Your assessments as evidence of skills

**Assessment 1:** survey data analysis in R, data visualisation with ggplot2, and communication to non-specialists.

**Assessment 2:** linear regression in R, causal inference from experiments, and statistical reporting.

At the guest event, ask how professionals use these skills. Afterwards, note one example from your own work that you could explain in an interview.

# Course expectations

**Regular practice, curiosity, and asking for help matter more than arriving with experience.**

## Before class

Check **Canvas modules** for readings, preparation, and announcements. Use your personal timetable for the session time and location.

## During class

Take part in the questions and activities. In workshops, work through the practical tasks at your own pace and ask your tutor when you get stuck. Discuss ideas respectfully and give others room to contribute.

## Between sessions

Revisit the code. Run it, change one thing, and explain what the output means. Save your scripts and note what remains unclear.

## Before asking for feedback

Bring a specific question and the relevant data, code, output, or draft. Explain what you tried and where the difficulty begins.

## When submitting work

Acknowledge your sources and represent evidence honestly. Follow the current University and assessment guidance on academic integrity and permitted use of tools.

### Get help early

Questions about the module: **teaching team on Canvas**.
Software or access: **Student IT support**.
Study support or mitigating circumstances: **Student Support Hub**.

# Course overview

**Lectures run in Weeks 1–10. Workshops run in Weeks 2–11, so the workshop number is one behind the teaching week.**

Select a week or its page number to open its guide. Use your personal timetable for dates, times, and rooms.


| Week | Lecture | Workshop | Page |
|---|---|---|---|
| 1 | Getting started | No workshop | 9 |
| 2 | Surveys and public opinion | 1. Getting started with data and R | 10 |
| 3 | Measurement and trends | 2. Prepare data | 11 |
| 4 | Groups and descriptive statistics | 3. Analyse data | 12 |
| 5 | Visualisation and data journalism | 4. Visualise data | 13 |
| 6 | Experiments and causation | 5. Assessment 1 support | 14 |
| 7 | Treatment effects | 6. Describe experimental data | 15 |
| 8 | Regression | 7. Estimate treatment effects | 16 |
| 9 | Statistical uncertainty | 8. Interpret statistical evidence | 17 |
| 10 | Skills and careers | 9. Communicate research findings | 18 |
| 11 | No lecture | 10. Assessment 2 support | 19 |


**Assessment support:** Workshop 5 in Week 6 supports the data journalism article. Workshop 10 in Week 11 supports the experimental report.

# Week 1: Getting started

Why use numbers to understand politics?

Lecture 1 · No workshop

## Before class

Check your timetable and open the course area. Think of a political claim you would like to investigate.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 1: Welcome and introduction

We begin with political questions and the role of evidence in answering them. We consider what quantitative data can reveal, what they can miss, and how the module's two parts fit together.

**Lecture materials · Materials to follow**

## No workshop this week

Use the time to find the course on Canvas and familiarise yourself with the materials.

## Take away

A political question and a first idea of the evidence you would need.

**Check your understanding:** Identify a political claim you would like to investigate. What evidence would help you assess it?

# Week 2: Surveys and public opinion

How do we know what people think?

Lecture 2 · Workshop 1

## Before class

Make sure you can open RStudio Desktop. If you cannot access it, ask for help before the workshop.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 2: Surveys, samples, and political evidence

We explore how surveys turn individual responses into evidence about public opinion. We ask how samples are recruited, what it means for a sample to be representative, and where to find political data.

Lecture materials · Materials to follow

## Workshop 1: Getting started with data and R

Read a small table of YouGov results, make a simple calculation in R, and save your first script.

Workshop materials · Materials to follow

## Take away

A saved R script containing your first calculation.

**Check your understanding:** Describe what a survey result measures and whose views it is intended to represent.

# Week 3: Measurement and trends

How do we know whether opinion is changing?

Lecture 3 · Workshop 2

## Before class

Reopen the R script you saved in Workshop 1 and try running it again.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 3: Measurement and trends

We consider how political attitudes are measured, how a concept becomes a variable, and how comparisons over time can help us identify change. We also ask when changes in wording or measurement complicate a comparison.

**Lecture materials · Materials to follow**

## Workshop 2: Prepare data

Begin finding and importing data, then inspect and prepare it in R. Practise selecting, filtering, and cleaning data for analysis.

**Workshop materials · Materials to follow**

## Take away

A script that imports and prepares data for analysis.

**Check your understanding:** Check that observations being compared measure the same thing in a meaningful way.

# Week 4: Groups and descriptive statistics

Who thinks what?

Lecture 4 · Workshop 3

## Before class

Review the data-preparation steps from the previous workshop and note anything you want to revisit.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 4: Groups and descriptive statistics

We use descriptive statistics to explore differences between groups. We consider proportions and other summaries, and ask how to describe a pattern without claiming more than the data support.

**Lecture materials · Materials to follow**

## Workshop 3: Analyse data

Summarise political data in R, compare groups or periods, and identify a finding worth explaining.

**Workshop materials · Materials to follow**

## Take away

A descriptive summary and a clearly stated finding.

**Check your understanding:** Describe a difference in plain language, including the groups and units being compared.

# Week 5: Visualisation and data journalism

What makes a visualisation trustworthy?

Lecture 5 · Workshop 4

## Before class

Bring a finding from your practice analysis that you could explain in a chart.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 5: Visualisation and data journalism

We examine how design choices shape the way readers understand evidence: labels, scales, colour, and graphical integrity all matter. The second part introduces Assessment 1 and how to turn a finding into a clear article.

**Lecture materials · Materials to follow**

## Workshop 4: Visualise data

Create, refine, label, and export a chart using ggplot2.

**Workshop materials · Materials to follow**

## Take away

An exported chart with readable labels and a source credit.

**Check your understanding:** Choose a visual form that makes the comparison clear and gives readers the information they need.

# Week 6: Experiments and causation

How can an experiment tell us what causes what?

Lecture 6 · Workshop 5

## Before class

Read the Assessment 1 brief. Bring your question, data, code, or a draft figure to Workshop 5.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 6: Experiments and causal questions

We introduce counterfactual thinking, treatment and control groups, and random assignment. We ask why association alone may not tell us what caused an outcome, and how an experiment can help.

**Lecture materials · Materials to follow**

## Workshop 5: Assessment 1 support

Bring your question, data, code, or a draft figure. Use the session to develop your article and troubleshoot with your tutor.

**Workshop materials · Materials to follow**

## Take away

A clearer next step for your data journalism article.

**Check your understanding:** Explain the difference between a descriptive pattern and a causal claim.

# Week 7: Treatment effects

How do we measure an effect?

Lecture 7 · Workshop 6

## Before class

Revisit the distinction between treatment and control, and think about what the experiment measures.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 7: Comparing treatment and control

We define outcomes and treatment effects, compare group means, and consider why an estimated effect can differ across samples. We also ask whether the size of a difference matters in political terms.

**Lecture materials · Materials to follow**

## Workshop 6: Describe experimental data

Identify the treatment and outcome, calculate group means, and compare the results.

**Workshop materials · Materials to follow**

## Take away

Treatment and control group means and their difference.

**Check your understanding:** Interpret a difference between treatment and control groups in the units of the outcome.

# Week 8: Regression

Why use regression to analyse experiments?

Lecture 8 · Workshop 7

## Before class

Review your treatment and control group summaries from Workshop 6.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 8: From group differences to regression

We introduce simple linear regression as a way to estimate an experimental treatment effect. We interpret the intercept and treatment coefficient and connect the model to a difference in group means.

**Lecture materials · Materials to follow**

## Workshop 7: Estimate treatment effects

Fit a simple model using lm() in R and interpret its coefficients.

**Workshop materials · Materials to follow**

## Take away

A regression estimate and an explanation of its treatment coefficient.

**Check your understanding:** Explain what a treatment coefficient says about the difference between groups.

# Week 9: Statistical uncertainty

How certain are we about an effect?

Lecture 9 · Workshop 8

## Before class

Reopen your regression output and identify the intercept and treatment coefficient.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 9: Uncertainty and differences in effects

We interpret confidence intervals and p-values alongside the size and direction of an estimate. We also consider why an average effect may not describe every group, and why subgroup claims need careful evidence.

**Lecture materials · Materials to follow**

## Workshop 8: Interpret statistical evidence

Extract estimates and measures of uncertainty and practise explaining them; an interaction exercise may be offered as an extension.

**Workshop materials · Materials to follow**

## Take away

An interpretation of an estimate, its confidence interval, and p-value.

**Check your understanding:** Describe an estimate and its uncertainty without treating statistical significance as the whole story.

# Week 10: Skills and careers

Where can these skills take you?

Lecture 10 · Workshop 9

## Before class

Prepare a question for the guest speakers. Bring your experimental results to the workshop.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

## Lecture 10: Guest event and professional practice

Hear from professionals about how they use quantitative and analytical skills. Reflect on the questions they ask, the evidence they use, and the ways they communicate their findings.

**Lecture materials · Materials to follow**

## Workshop 9: Communicate research findings

Prepare a results table or figure, document your analysis, and practise writing up an experimental result.

**Workshop materials · Materials to follow**

## Take away

A results table or figure and a short explanation of the finding.

**Check your understanding:** Connect a skill you have practised to an example you could discuss in an application or interview.

# Week 11: Assessment 2 support

How do I bring my analysis together?

No lecture · Workshop 10

## Before class

Read the Assessment 2 brief. Bring your script, results, or draft, plus a specific question for your tutor.

**Weekly readings and any required preparation: check Canvas. Readings are still being finalised.**

There is no lecture this week. Use the time to bring together your question, literature, analysis, and interpretation.

## Workshop 10: Assessment 2 support

Bring your data, script, results, or draft and use the session to work on the experimental report with help from your tutor.

**Workshop materials · Materials to follow**

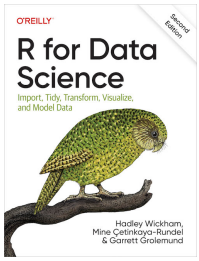
## Take away

A clear plan for the remaining revisions to your experimental report.

**Check your understanding:** Check that your report answers its question, interprets uncertainty, and explains the study's limitations.

# What's next?

**Want to learn more about data science and causal inference? Build on what you've learnt on Making Sense of Politics by reading these excellent books — all free online.**

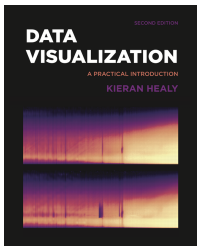



## R for Data Science

Hadley Wickham, Mine Çetinkaya-Rundel, and Garrett Grolemund

Build on your R skills as you learn to import, tidy, and visualise data, ready to explore questions of your own.

Click to read for free online

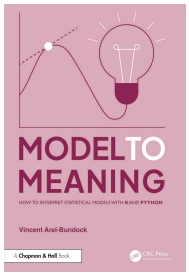



## Data Visualization: A Practical Introduction

Kieran Healy

Develop your ggplot2 skills as you discover what makes a great figure and how you can communicate your findings with a wider audience.

Click to read for free online

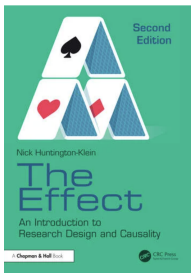



## Model to Meaning

Vincent Arel-Bundock

Learn how to model data using machine learning and statistics, and turn their results into meaningful answers.

Click to read for free online




## The Effect

Nick Huntington-Klein

Explore how research design helps us uncover cause and effect, and learn how to judge what evidence can tell us about the world.

Click to read for free online